

Improving Survival Prediction in NSCLC through Clinical Report Embeddings: A Comparison with Structured Tabular Data and Multimodal Fusion Model

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APOLLO 11 - Istituto Nazionale dei Tumori

- Overview:** Apollo 11 (NCT05550961) is a 3-year Italian project involving **48 hospitals**, led by **Fondazione IRCCS Istituto Nazionale dei Tumori, Milan**.
- Goal:** Integrate **real-world clinical, molecular, and imaging data** from patients with advanced lung cancer treated with immunotherapy.
- Innovation:** A shared **virtual biobank** enabling **AI-based predictive modeling** and large-scale translational research.



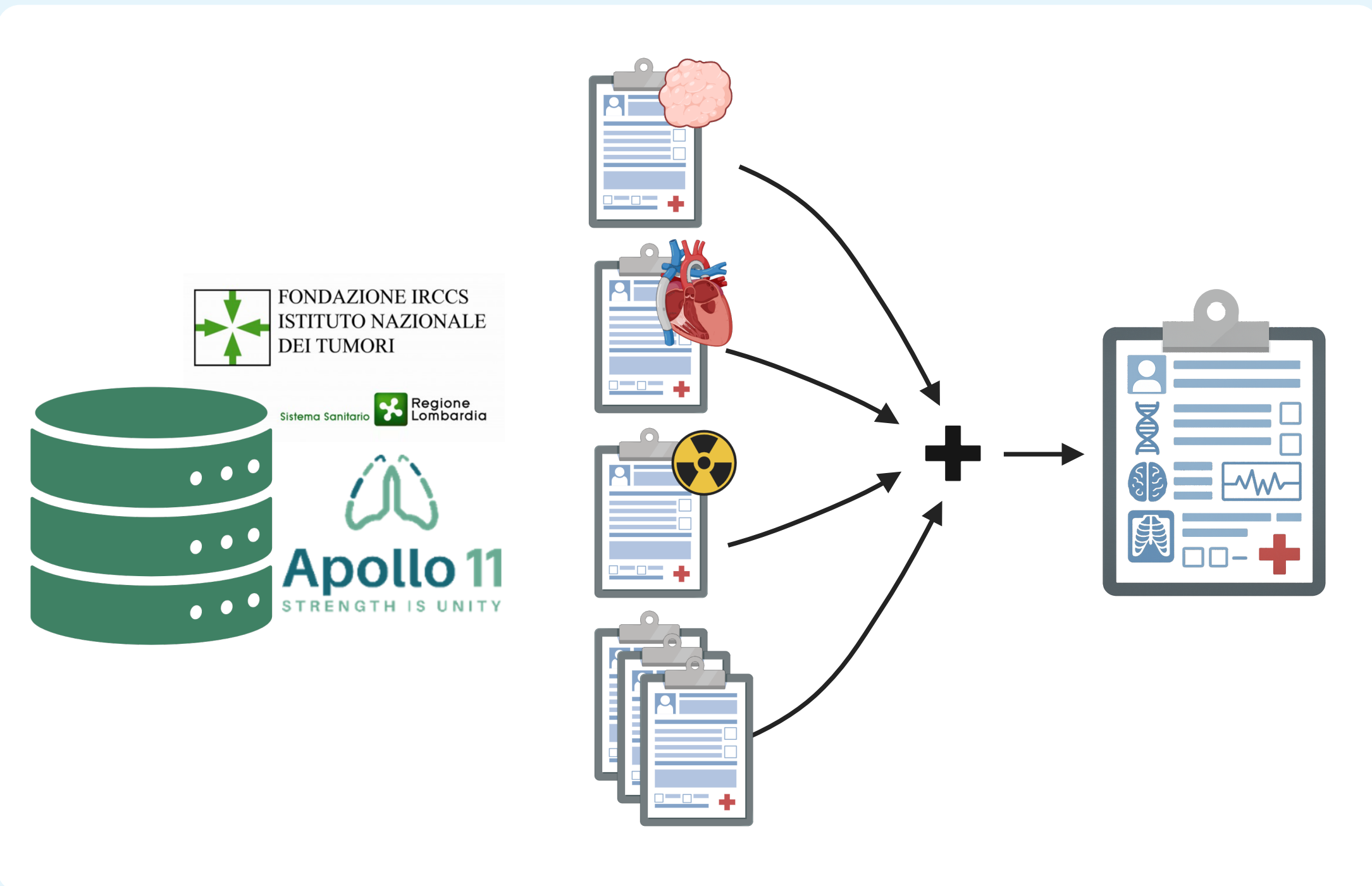
- Challenge:** Predicting **immunotherapy response in NSCLC** remains difficult due to data fragmentation.
- APOLLO 11 solution:** A unified infrastructure supporting **AI-driven, multicenter analyses** of real-world data.
- Opportunity:** The platform enables **AI-driven predictive models** for better patient outcomes.

INTRODUCTION

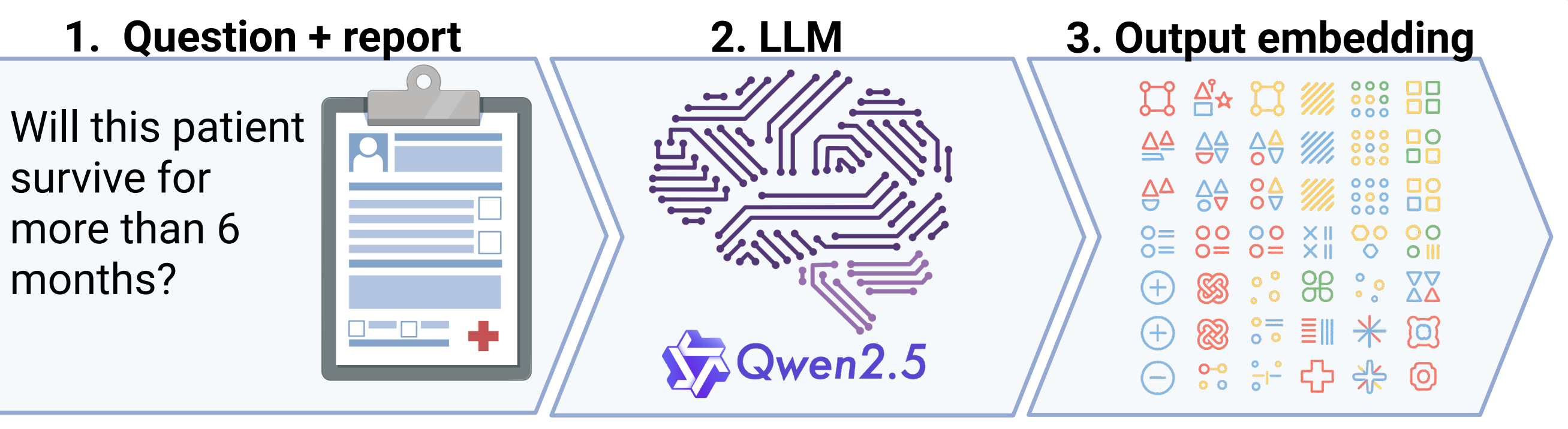
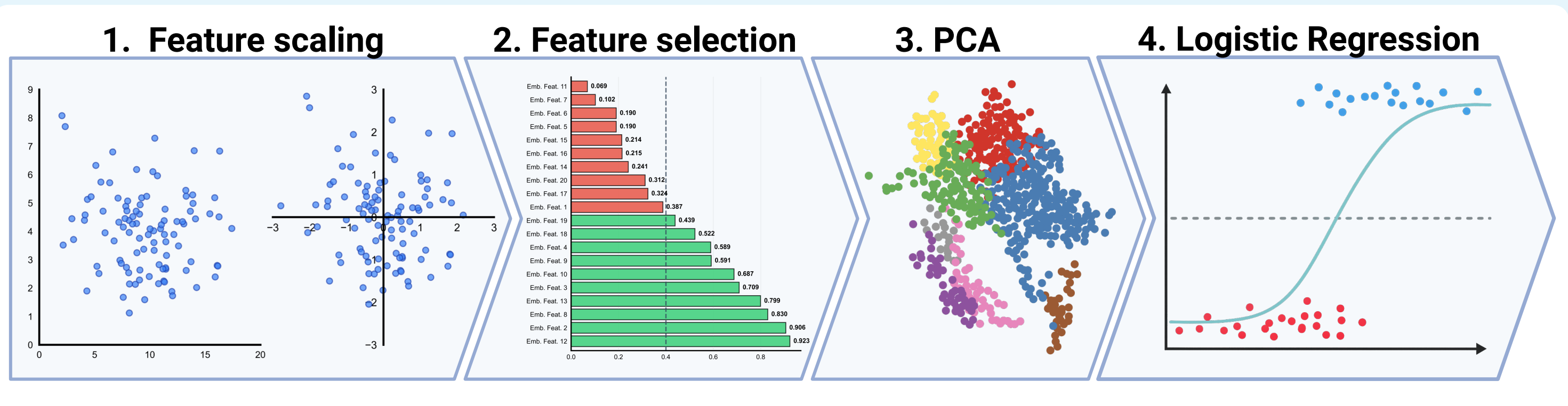
- Problem:**
 - Predictive modeling in NSCLC is limited by the difficulty in **extracting exhaustive features** from clinical records.
 - The comprehensive use of **entire clinical reports** for predictive modeling remains largely **unexplored**.
- Approach:**
 - Extract **embeddings** from **clinical reports** using a large language model (LLM).
 - Train a classifier for **6-months overall survival** and compare it with one trained on **tabular data**.
- Objective:**
 - Compare** the models trained on **structured** and **unstructured** data
 - Test the **complementarity** of **embeddings** and **tabular data** with a **multimodal fusion model**.

METHODS

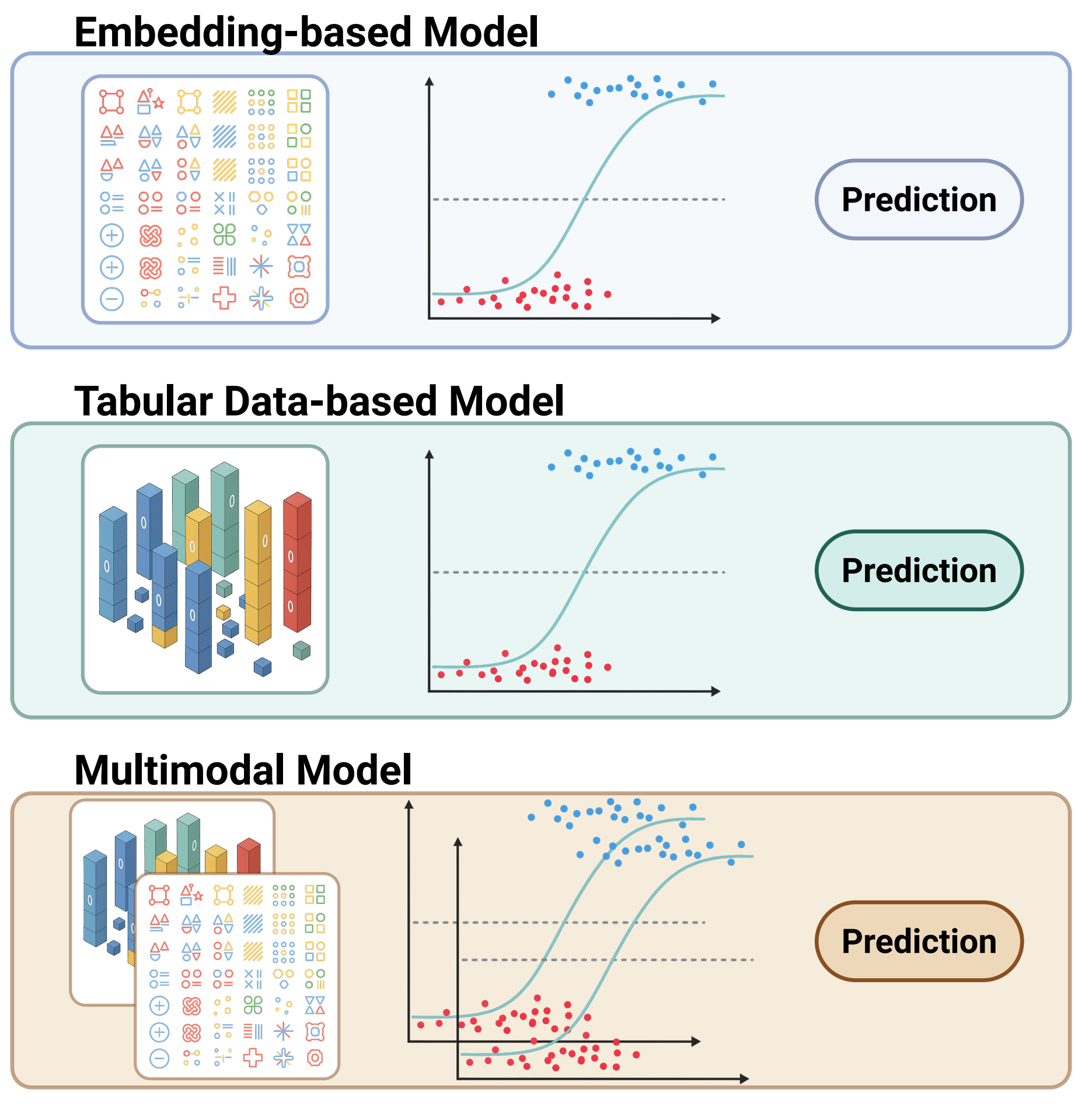
- Retrospective cohort: **633 NSCLC patients** from **Istituto Nazionale dei Tumori** who participated in the **Apollo 11 project**.
- Inclusion criteria: **Clinical reports available prior to the start of immunotherapy**, covering various specialties (**oncology, cardiology, etc.**).
- Data aggregation: All relevant reports for each patient were **combined into a single, comprehensive document**.



- Embedding Processing:**
 - Feature scaling applied to **normalize data**.
 - Feature selection using **SelectKBest**, retaining **182 features**.
 - Dimensionality reduction via **PCA**, reducing to **41 features**.
- Classifier:**
 - Logistic Regression** trained on the embeddings, with a training set of **541 patients** and a test set of **92 patients**.
- Evaluation:**
 - Evaluation metrics: **AUC, F1 macro, Sensitivity, and Specificity**
 - Evaluation sets: **train set, test set** and **5-fold cross validation**.



- Model:** **Qwen2.5-7B-Instruct-1M** with 4-bit quantization to reduce memory while preserving performance.
- Prompt design:** Each clinical report was preceded by a **task-specific prompt** - "Question: Will the patient survive for more than 6 months? Report:" - to orient the embedding extraction toward survival prediction.
- Embedding extraction:** **Final-layer hidden states** were taken from the model's forward pass, with **prompt tokens masked out**. Remaining tokens were **averaged** and **L2-normalized** to obtain a single **3584-dimensional embedding** per patient.
- Long-report handling:** For long reports, a **token-aware word-level chunking** strategy was used; chunk embeddings were **mean-pooled** and **L2-normalized** to produce the final representation.



- Models Comparison:**
 - Embedding-based** classifier
 - Tabular** data-based classifier
 - Multimodal** classifier (embedding + tabular)

RESULTS

Train	Metric	Embedding-based	Tabular	Data-based	Multimodal
	AUC	0.89	0.74		0.90
	F1	0.78	0.67		0.78
	Precision	0.78	0.67		0.78
	Sensitivity	0.76	0.70		0.76
	Specificity	0.84	0.68		0.84

Test	Metric	Embedding-based	Tabular	Data-based	Multimodal
	AUC	0.81	0.70		0.83
	F1	0.74	0.66		0.74
	Precision	0.74	0.66		0.74
	Sensitivity	0.73	0.63		0.73
	Specificity	0.79	0.73		0.79

5-fold CV	Metric	Embedding-based	Tabular	Data-based	Multimodal
	AUC	0.86 ± 0.03	0.72 ± 0.04		0.86 ± 0.03
	F1	0.76 ± 0.04	0.68 ± 0.04		0.76 ± 0.04
	Precision	0.76 ± 0.03	0.67 ± 0.04		0.75 ± 0.04
	Sensitivity	0.75 ± 0.05	0.71 ± 0.05		0.75 ± 0.04
	Specificity	0.81 ± 0.06	0.67 ± 0.07		0.81 ± 0.06

- Embedding-based model** shows the best overall performance, with strong generalization and balanced sensitivity and specificity.
- Tabular model** performs worse, highlighting limited information in structured data alone.
- Multimodal model** performs similarly to embedding-based, indicating **no added value** from tabular data.

CONCLUSIONS

- The **embedding-based model** outperforms traditional tabular approaches, capturing **latent clinical information** that structured variables fail to represent.
- Text-derived embeddings** from clinical reports provide a richer and more comprehensive description of patient status.
- The **multimodal model** performs similarly, suggesting that most predictive signal resides in the textual data.
- These results emphasize the **value of unstructured clinical text** for outcome prediction in oncology.

CONTACT INFORMATION



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